Linear ODEs

Definition (Classification of ODEs)

A *linear ODE* (ordinary differential equation) is a linear combination of derivatives of an unknown function $y(t)$ whose coefficients may be functions of t but not y:

$$f_n(t)y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

- The ODE is of order n if $f_n(t) \neq 0$.
- If $g(t) = 0$ then the ODE is *homogeneous*.
- If $g(t) \neq 0$ then the ODE is *inhomogeneous*.

Exercise 1.1: Classify the following ODEs

1. $y''(t) + \cos(t)y'(t) = t \sin t$

   Linear, second-order, inhomogeneous

2. $y'''(t) - y(t)^2 = \mathrm{e}^t \log y(t)$

   Non-linear

Initial value problems (IVPs)

Definition

1. Solutions to ODEs contain unknown constants (solutions are non-unique).
2. Constants can be fixed by specifying *initial values* at $t = t_0$ (usually $t_0 = 0$).
3. ODEs of order n require n initial values.

$$\text{Usually } y(t_0) = y_0,\ y'(t_0) = y'_0,\ \ldots \text{ and } y^{(n-1)}(t_0) = y_0^{(n-1)}.$$

Exercise 1.2: Decide if the following are IVPs

1. $y''(t) = -3y(t)$ with $y(1) = 3$ and $y'(1) = 1$

   Initial value problem

2. $y''(t) = -3y(t)$ with $y(0) = 1$ and $y(2) = 0$

   Not an initial value problem

   This is a boundary value problem, which we will meet later

Integrating factor (revision)

ODEs of the form $y'(t) + f(t)y(t) = g(t)$ can be solved using an integrating factor:

$$I(t) = \mathrm{e}^{\int f(t)\mathrm{d}t}$$

Example: Solve

$$y'(t) - y(t) = 0$$

$$I(t) = \mathrm{e}^{\int(-1)\mathrm{d}t} = \mathrm{e}^{-t} \quad \Rightarrow \quad \Big(I(t)\,y(t)\Big)' = I(t)\cdot 0$$

$$\Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{e}^{-t}y) = 0 \quad \Rightarrow \quad \mathrm{e}^{-t}y = C_0 \quad \Rightarrow \quad y = C_0\mathrm{e}^{t}$$

Exercise 1.3: Find the integrating factor for the ODE

$$y'(t) - \frac{1}{t}y(t) = 1$$

In this case

$$f(t) = -\frac{1}{t}$$

and thus

$$I(t) = \mathrm{e}^{\int -\frac{1}{t}\mathrm{d}t} = \mathrm{e}^{-\ln(t)} = \mathrm{e}^{\ln(1/t)} = \frac{1}{t}$$

We now solve the problem using the IF by writing

$$\frac{\mathrm{d}}{\mathrm{d}t}[\frac{1}{t}y(t)] = \frac{1}{t}$$

and integrating both sides to give

$$\frac{1}{t}y(t) = \int \frac{1}{t}\mathrm{d}t = \log t + c \qquad \Rightarrow \qquad y(t) = t\log t + Ct$$

Lecture 2

- In Lecture 1, we defined linear ODEs+IVPs and revised integrating factors

Definition (Linear ODEs)

$$y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

Definition (Initial value problems, IVPs)

$$\text{Usually } y(t_0) = y_0,\ y'(t_0) = y_0',\ \ldots \text{ and } y^{(n-1)}(t_0) = y_0^{(n-1)}$$

- In Lecture 2, we will:
  1. Form a connection between 1st-order ODEs and linear algebra
  2. Express ODE solutions in terms of complementary and particular solutions
  3. Extend concepts to 2nd-order ODEs
  4. Learn how to use the reduction of order method

Connection to linear algebra

- Linear algebra: $\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$ where $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{w} \in \mathbb{R}^m$.
- $\mathbf{A} : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation:

$$\mathbf{A}(\boldsymbol{u}+\boldsymbol{v}) = \mathbf{A}\boldsymbol{u} + \mathbf{A}\boldsymbol{v} \quad \text{and} \quad \mathbf{A}(k\boldsymbol{u}) = k\mathbf{A}\boldsymbol{u} \quad \text{where} \quad \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n;\ k \in \mathbb{R}$$

- $\frac{\mathrm{d}}{\mathrm{d}t}$ and $f(t)\cdot$ are also linear transformations but they operate on functions.*
- $\mathsf{T} = \frac{\mathrm{d}}{\mathrm{d}t} + f(t)\cdot$ is a *linear transformation*

$$(\mathsf{T}y)(t) = \frac{\mathrm{d}y}{\mathrm{d}t} + f(t)y(t)$$

...you should check the definition is satisfied, i.e.,

$$\mathsf{T}(x+y) = \mathsf{T}x + \mathsf{T}y \quad \text{and} \quad \mathsf{T}(k\,y) = k\,\mathsf{T}y$$

*In Topic 4 we will extend the idea of these finite linear combinations to an infinite series. If you enjoy this aspect of the course then take MAST30026 Metric & Hilbert Spaces to gain a more detailed understanding.

We have the following connections

$\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$	$\leftrightarrow$	$y'(t) + f(t)y(t) = g(t)$
$\boldsymbol{v}$	$\leftrightarrow$	$y(t)$
$\boldsymbol{w}$	$\leftrightarrow$	$g(t)$
$\mathbf{A}$	$\leftrightarrow$	$\frac{\mathrm{d}}{\mathrm{d}t} + f(t)\cdot$

How to solve $\mathbf{A}\boldsymbol{v} = \boldsymbol{w}$?

- Solve $\mathbf{A}\boldsymbol{v}_c = \mathbf{0}$ to find complementary solutions.
  - Complementary solutions form a vector space, i.e., the solution space.
- Find a single solution to $\mathbf{A}\boldsymbol{v}_p = \boldsymbol{w}$, i.e., a particular solution.
- General solution: $\boldsymbol{v} = \boldsymbol{v}_c + \boldsymbol{v}_p$.

The same procedure works for linear ODEs

To solve $\mathsf{T}y = g$

- Solve $\mathsf{T}y_c = 0$ to find complementary solutions
  - Complementary solutions form a solution space.
- Find a (single) particular solution to $\mathsf{T}y_p = g$
- General solution: $y = y_c + y_p$

Exercise 1.4: Recall previous exercise

We considered the linear, first-order, non-homogeneous DE, using an IF.

$$y'(t) - \frac{1}{t}y(t) = 1 \quad \Rightarrow \quad y(t) = t\ln t + Ct$$

Find:

- Complementary solutions: $y_c = Ct$ (solution space is one-dimensional)
- Particular solution: $y_p = t\ln t$
- General solution: $y = Ct + t\ln t$

Second order linear ODEs

- Using notation from the linear algebra analogy, consider ODEs of form

$$(\mathsf{T}y)(t) = g(t) \quad \text{where} \quad \mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

- To find solutions we will use

Topic 1: Trial solutions, reduction of order, matrix diagonalisation

Topic 2: Power series

Topic 3: Laplace transforms

Topic 4: Fourier series

Topic 6: Fourier transforms

Trial solutions for constant coefficient ODEs (revision)

Consider $\mathsf{T}y = 0$, where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + b\frac{\mathrm{d}}{\mathrm{d}t} + c \quad \text{such that} \quad b, c \in \mathbb{R}$$

To solve, recall the approach:

- Trial a solution of the form $y(t) = \mathrm{e}^{\lambda t}$, where $\lambda \in \mathbb{C}$.
- This leads to the characteristic equation $\lambda^2 + b\lambda + c = 0$
- Then $\lambda_\pm = \dfrac{-b \pm \sqrt{b^2 - 4c}}{2}$
- Therefore, $y = C_+\mathrm{e}^{\lambda_+ t} + C_-\mathrm{e}^{\lambda_- t}$ if $b^2 - 4c \neq 0$ so that $\lambda_+ \neq \lambda_-$

Reduction of order

Solve $\mathsf{T}y = 0$ for the case where $c = \dfrac{b^2}{4} \Rightarrow b^2 - 4c = 0$

- Method of trial solution only gives one solution, $y(t) = \mathrm{e}^{-\frac{b}{2}t}$
- Trial solution of the form $y(t) = u(t)\mathrm{e}^{-\frac{b}{2}t}$
- $y' = \left(u' - \dfrac{b}{2}u\right)\mathrm{e}^{-\frac{b}{2}t}$ and $y'' = \left(u'' - bu' + \dfrac{b^2}{4}u\right)\mathrm{e}^{-\frac{b}{2}t}$
- $\mathsf{T}y = \Big(u'' + u'(-b+b) + u(b^2/4 - b^2/2 + b^2/4)\Big)\mathrm{e}^{-\frac{b}{2}t} = 0$
- $u'' = 0$, so $u = C_1 t + C_2$ and hence we have found the general solution

$$y(t) = (C_1 t + C_2)\,\mathrm{e}^{-\frac{b}{2}t}$$

Reduction of order—general theory

Definition

Let $v(t)$ be a non-zero solution of $\mathsf{T}v = 0$ where T is an arbitrary second-order linear operator, then the general solution to $\mathsf{T}y = g$ can be found in the form $y(t) = u(t)v(t)$.

- Chain rule gives $y' = u'v + uv'$ and $y'' = u''v + 2u'v' + uv''$
- Therefore

$$\begin{aligned} \mathsf{T}y &= u''v + 2u'v' + uv'' + f_1(u'v + uv') + f_0uv \\ &= u\,\mathsf{T}v + u''v + 2u'v' + f_1u'v \end{aligned}$$

- $\mathsf{T}v = 0$ and hence $\mathsf{T}y = u''v + 2u'v' + f_1u'v = g$. <u>No f_0 (explicitly)</u>
- Rearranging gives $u'' + \left(\dfrac{2v'}{v} + f_1\right)u' = \dfrac{g}{v}$. <u>What do you notice?</u>

- Let $w = u'$, so $w' + \left(\frac{2v'}{v} + f_1\right) w = \frac{g}{v}$

- We have *reduced* the order of our ODE from two to one.

- Solve using an integrating factor:

$$I(t) = \mathrm{e}^{\int \left(\frac{2v'}{v} + f_1\right)\mathrm{d}t} = \mathrm{e}^{2\log v + \int f_1 \mathrm{d}t} = \mathrm{e}^{\log v^2}\, \mathrm{e}^{\int f_1 \mathrm{d}t} = v^2 \mathrm{e}^{\int f_1 \mathrm{d}t}$$

- Thus

$$\frac{\mathrm{d}}{\mathrm{d}t}[Iw] = \frac{Ig}{v} \quad \Rightarrow \quad w = \frac{1}{I}\int \frac{Ig}{v}\mathrm{d}t$$

$$\Rightarrow \quad u = \int \left[\frac{1}{I}\int \frac{Ig}{v}\mathrm{d}t\right]\mathrm{d}t \quad \Rightarrow \quad y = v\int \left[\frac{1}{I}\int \frac{Ig}{v}\mathrm{d}t\right]\mathrm{d}t$$

Note: the two integrals will give us the required two unknown constants.

Summary of reduction of order

1. Given known complementary solution, $v(t)$, to a second order ODE, write general solution as $y(t) = u(t)v(t)$

2. This reduces the order of the ODE to one: $w' + \left(\frac{2v'}{v} + f_1\right) w = \frac{g}{v}$ where $w = u'$

3. Use the integrating factor $I = v^2 \mathrm{e}^{\int f_1 \mathrm{d}t}$ to solve for w.

4. This leads to the general solution $y = v \int \left[\frac{1}{I} \int \frac{Ig}{v} \mathrm{d}t\right] \mathrm{d}t$

Finding the first solution for $\mathsf{T}v = 0$ *is usually the difficult task.*

Exercise 1.5: Apply reduction of order to the ODE

$$y'' + \frac{1}{t}y' - \frac{1}{t^2}y = g(t) \quad \text{given that } y = t \text{ is a complementary solution}$$

We write the solution as $y(t) = v(t)\,u(t) = t\,u(t)$.

We therefore have $y' = (t\,u)' = tu' + u$ and $y'' = (t\,u' + u)' = tu'' + 2u'$.

Reduction of order gives

$$w' + \frac{3}{t}w = \frac{g}{t} \quad \text{where} \quad w = u'$$

Integrating factor is $I(t) = v^2 \mathrm{e}^{\int t^{-1}} = t^2 \mathrm{e}^{\ln(t)} = t^2 \times t = t^3 \quad \Rightarrow \quad w = \dfrac{1}{t^3}\displaystyle\int t^2\, g\ \mathrm{d}t$

Find the general solution when $g = 0$

$$t^3 w = \int t^2 \cdot 0 \, \mathrm{d}t = C_1 \qquad \Rightarrow \qquad w = u' = \frac{C_1}{t^3} \qquad \Rightarrow \qquad u = -\frac{C_1}{2t^2} + C_2$$

Therefore

$$y = t\,u = -\frac{C_1}{2t} + C_2 t$$

Find the general solution when $g = \mathrm{e}^t$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(t^3 w\right) = t^2 \mathrm{e}^t$$

$$t^3 w = \int t^2 \mathrm{e}^t \mathrm{d}t = t^2 \mathrm{e}^t - \int 2t\mathrm{e}^t \mathrm{d}t + C_1$$

$$t^3 w = t^2 \mathrm{e}^t - 2t\mathrm{e}^t + \int 2\mathrm{e}^t \mathrm{d}t + C_1 = (t^2 - 2t + 2)\mathrm{e}^t + C_1$$

$$(w =)u' = \frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3} + \frac{C_1}{t^3}$$

$$u = \int \left\{\frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3}\right\} \mathrm{d}t - \frac{C_1}{2t^2} + C_2$$

$$= \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2} - \frac{C_1}{2t^2} + C_2$$

$$y = t\left(\frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2} - \frac{C_1}{2t^2} + C_2\right) = \left(1 - \frac{1}{t}\right)\mathrm{e}^t - \frac{C_1}{2t} + C_2 t$$

Workings

$$
\begin{aligned}
\int \left\{ \frac{\mathrm{e}^t}{t} - \frac{2\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t^3} \right\} \mathrm{d}t &= \int \frac{\mathrm{e}^t}{t} \mathrm{d}t - 2\left(-\frac{\mathrm{e}^t}{t} + \int \frac{\mathrm{e}^t}{t} \mathrm{d}t \right) + 2\left(-\frac{\mathrm{e}^t}{2t^2} + \int \frac{\mathrm{e}^t}{2t^2} \mathrm{d}t \right) \\
&= -\frac{\mathrm{e}^t}{t^2} + \frac{2\mathrm{e}^t}{t} - \int \frac{\mathrm{e}^t}{t} \mathrm{d}t + \left(-\frac{\mathrm{e}^t}{t} + \int \frac{\mathrm{e}^t}{t} \mathrm{d}t \right) \\
&= \frac{\mathrm{e}^t}{t} - \frac{\mathrm{e}^t}{t^2}
\end{aligned}
$$

Existence and uniqueness of solutions

Theorem

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

If f_0, f_1 and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is a linear sum of 2 linearly independent complementary solutions and a single particular solution.
  - Solutions space (i.e., all complementary solutions) is a 2-dimensional space.
- IVPs: When supplemented with initial values $y(t_0)$ and $y'(t_0)$ where $t_0 \in D$, it has a unique solution which is twice differentiable.

Linear independence

Linear algebra:

Two vectors, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, are linearly dependent if there exists c such that $\boldsymbol{v}_1 = c\,\boldsymbol{v}_2$.

Differential equations:

Two functions, $y_1(t)$ and $y_2(t)$, linearly dependent if there exists c such that

$$y_1(t) = c\,y_2(t) \quad \text{for all} \quad t.$$

If not, then they are linearly independent

Exercise 1.6: Are $y_1 = \mathrm{e}^{-2t}$ and $y_2 = 2\mathrm{e}^{-2t}$ linearly independent.

No, because $y_1 = \dfrac{1}{2} y_2$

The Wronskian

Definition (The Wronskian)

Given two functions y_1 and y_2, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2](t) = \det \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

- y_1 and y_2 are linearly independent if

$$\mathrm{wr}[y_1, y_2](t) \neq 0 \quad \text{for some } t$$

  - y_1 and y_2 are linearly dependent if $\mathrm{wr}[y_1, y_2](t) = 0$ for all t
  - $\mathrm{wr}[y_1, y_2] = 0$ for some t tells you nothing

Theorem (Abel's differential equation identity)

Let y_1 and y_2 be two solutions of $y'' + f_1 y' + f_0 y = 0$. Then

$$\mathrm{wr}[y_1, y_2](t) = \mathrm{wr}[y_1, y_2](t_0)\mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau}$$

- If the Wronskian is non-zero anywhere it will be non-zero everywhere.
- If the Wronskian is zero anywhere it will be zero everywhere.

If we know one solution, say y_1, we can use Abel's identity and Slide 5 to find the second linearly independent solution, y_2, by solving the ODE

$$y_1 y_2' - y_1' y_2 = \mathrm{wr}[y_1, y_2](t_0)\mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau}$$

This is equivalent to using reduction of order.

Exercise 1.7: Study linear independence of solutions to below ODE

$$y'' + by' + \frac{b^2}{4}y = 0 \quad \Rightarrow \quad y(t) = C_1\,\mathrm{e}^{-\frac{b}{2}t} + C_2\,t\mathrm{e}^{-\frac{b}{2}t}$$

Therefore, study Wronskian of $\mathrm{e}^{-\frac{b}{2}t}$ and $t\,\mathrm{e}^{-\frac{b}{2}t}$, which is

$$\begin{aligned}
\mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](t) &= \det\begin{pmatrix} \mathrm{e}^{-\frac{b}{2}t} & t\mathrm{e}^{-\frac{b}{2}t} \\ -\frac{b}{2}\mathrm{e}^{-\frac{b}{2}t} & \left(1-\frac{b}{2}t\right)\mathrm{e}^{-\frac{b}{2}t} \end{pmatrix} \\
&= \left(1 - \frac{b}{2}t - t\left(-\frac{b}{2}\right)\right)\mathrm{e}^{-\frac{b}{2}t}\mathrm{e}^{-\frac{b}{2}t} \\
&= \mathrm{e}^{-bt}
\end{aligned}$$

which is non-zero for some t, e.g., $\mathrm{wr}(1) = \mathrm{e}^{-b}$.

Hence, these two solutions are linearly independent and are a basis for the solution space.

- Choose $t_0 = 0$
- Then $\mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](t_0) = \mathrm{e}^{-bt_0} = 1$
- Note $f_1 = b \Rightarrow \mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau} = \mathrm{e}^{-\int_0^t b\mathrm{d}\tau} = \mathrm{e}^{-bt}$
- Thus, $\mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](t) = \mathrm{wr}[\mathrm{e}^{-\frac{b}{2}t}, t\mathrm{e}^{-\frac{b}{2}t}](0)\mathrm{e}^{-\int_0^t f_1(\tau)\mathrm{d}\tau} = 1 \times \mathrm{e}^{-bt} = \mathrm{e}^{-bt}$
- Confirming Abel's identity

Exercise 1.8: Find solutions of the form $\phi = r^\lambda$ for the below ODE

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}\phi}{\mathrm{d}r} - \frac{n^2}{r^2}\phi = 0, \qquad r > 0, \qquad n \in \{0, 1, 2, \ldots\}^*$$

We "trial"

$$\phi = r^\lambda \quad \Rightarrow \quad \phi' = \lambda\, r^{\lambda-1} \quad \Rightarrow \quad \phi'' = \lambda(\lambda-1)\, r^{\lambda-2}$$

Substituting into the ODE

$$\lambda(\lambda-1)r^{\lambda-2} + \frac{1}{r}\lambda r^{\lambda-1} - \frac{n^2}{r^2}r^\lambda = 0$$

$$\lambda^2 - n^2 = 0$$

which has solution $\lambda = \pm n$.

*This is the Euler equation from, e.g., electrostatics, which we will see again in Topic 5

For $n \neq 0$:

The Wronskian shows r^n and r^{-n} are linearly independent (details left as exercise).

Hence the general solution is $\phi = C_1 r^n + C_2 r^{-n}$.

For $n = 0$:

We have only one solution $\phi_1 = \mathrm{e}^{\pm 0} = 1$.

The Wronskian is proportional to $\mathrm{e}^{-\int \frac{1}{r}\mathrm{d}r} = \frac{1}{r}$

Thus, from Abel's identity

$$\phi_2' = \frac{1}{r} \qquad \Rightarrow \qquad \phi_2 = \log r.$$

so the general solution for $n = 0$ is $\phi = C_1 + C_2 \log r$

Higher-order (linear) ODEs

Theorem (Solutions of an nth order linear ODE/IVP)

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^n}{\mathrm{d}t^n} + f_{n-1}(t)\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} + \ldots + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)\cdot$$

If $f_0, f_1, \ldots, f_{n-1}$ and g are all continuous over $D = [a, b]$, then over D:

- ODEs: The general solution to $\mathsf{T}y = g$ is the sum of n linearly independent complementary solutions and a single particular solution.
  - Solution space (i.e., all complementary solutions) form an n-dimensional vector space.
- IVPs: When supplemented with initial values $y(t_0), y'(t_0), \ldots, y^{(n-1)}(t_0)$ where $t_0 \in D$, it has a unique solution which is n-times differentiable.

Definition (The Wronskian)

Given n functions $y_1, y_2, \ldots, y_n$, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

- $y_1, y_2, \ldots, y_n$ are linearly independent if there exists a t such that

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) \neq 0$$

  - $y_1, y_2, \ldots, y_n$ are linearly dependent if $\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = 0$ for all t

Method of trial solutions (higher order)

Exercise 1.9: Find the general solution to the below ODE by trialing $y = \mathrm{e}^{\lambda t}$.

$$y''' + y'' - 4y' - 4y = 0$$

$y = \mathrm{e}^{\lambda t} \Rightarrow (\lambda + 1)(\lambda^2 - 4) = 0$. Thus $\lambda = -1, \pm 2$

Let $y_1 = \mathrm{e}^{-2\,t}$, $y_2 = \mathrm{e}^{-t}$, $y_3 = \mathrm{e}^{2\,t}$. The Wronskian is

$$\begin{aligned} \mathrm{wr}[y_1, y_2, y_3](t) &= \det \begin{pmatrix} \mathrm{e}^{-2t} & \mathrm{e}^{-t} & \mathrm{e}^{2t} \\ -2\mathrm{e}^{-2t} & -\mathrm{e}^{-t} & 2\mathrm{e}^{2t} \\ 4\mathrm{e}^{-2t} & \mathrm{e}^{-t} & 4\mathrm{e}^{2t} \end{pmatrix} \\ &= 12\mathrm{e}^{-t} \end{aligned}$$

and so the general solution is $y = C_1\mathrm{e}^{-2t} + C_2\mathrm{e}^{-t} + C_3\mathrm{e}^{2t}$.

Reduction of order also works in higher dimensions but we won't cover that in MAST20030.

Summary of Linear ODEs

1. An nth order linear ODE has n *linearly independent* complementary solutions and a single particular solution. It has a unique solution if n initial values are specified.

2. Check for linear independence using the Wronskian.

3. If one complementary solution for a 2nd order ODE is known, the general solution can be found through reduction of order.

Systems of first-order linear ODEs

Let's take a different perspective on linear ODEs.

Definition (Higher order ODEs as a system)

$$y^{(n)}(t) + f_{n-1}(t)y^{(n-1)}(t) + \cdots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

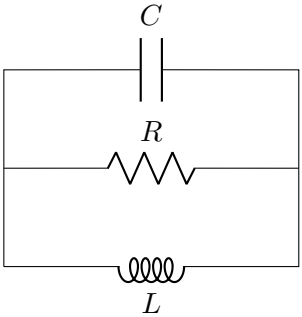
is equivalent to the following system of n first order ODEs:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} z_1(t) \\ z_2(t) \\ \vdots \\ z_{n-1}(t) \\ z_n(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -f_0(t) & -f_1(t) & \cdots & -f_{n-1}(t) \end{pmatrix}\begin{pmatrix} z_1(t) \\ z_2(t) \\ \vdots \\ z_{n-1}(t) \\ z_n(t) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ g(t) \end{pmatrix}$$

where $z_i(t) = y^{(i-1)}(t)$ for $i = 1, \ldots, n$

Case study 1: Tackling ventricular fibrillation (Problem 1.1)[1]

Express the governing system of ODEs for the following electrical circuit: (a) in matrix form and (b) as a single 2nd order ODE.



C, R and L are constants.

$$\left\{\begin{aligned} \frac{\mathrm{d}I}{\mathrm{d}t} &= \frac{V}{L}, \\ \frac{\mathrm{d}V}{\mathrm{d}t} &= -\frac{I}{C} - \frac{V}{RC} \end{aligned}\right\}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} I \\ V \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{pmatrix}\begin{pmatrix} I \\ V \end{pmatrix}$$

$$\frac{\mathrm{d}^2V}{\mathrm{d}t^2} + \left(\frac{1}{RC}\right)\frac{\mathrm{d}V}{\mathrm{d}t} + \left(\frac{1}{LC}\right)V = 0$$

[1] Anantharaman et al (2017) Role of peak current in conversion of patients with ventricular fibrillation. *Singapore Med J.*

Exercise 1.9: Write $y''(t) + c\,y = 0$ as a 1st-order ODE system

Start by writing $z(t) = y'(t) \Rightarrow z'(t) = y''(t)$

The 2nd-order ODE becomes a 1st-order ODE with two unknowns

$$z'(t) + c\,y(t) = 0$$

Reintroduce the definition of z, we recast our single 2nd-order ODE as

$$\begin{aligned} y'(t) &= z(t), \\ z'(t) &= -cy(t). \end{aligned}$$

In matrix form, it is $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$ or

$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} y(t) \\ z(t) \end{pmatrix}}_{\boldsymbol{y}'} = \underbrace{\begin{pmatrix} 0 & 1 \\ -c & 0 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} y(t) \\ z(t) \end{pmatrix}}_{\boldsymbol{y}}$$

Exercise 1.10: Write $y''(t) + b\,y'(t) + c\,y = 0$ as a 1st-order ODE system

The 2nd-order ODE becomes a 1st-order ODE with two unknowns

$$z'(t) + bz(t) + cy(t) = 0$$

Reintroduce the definition of z, we recast our single 2nd-order ODE as

$$\begin{aligned} y'(t) &= z(t), \\ z'(t) &= -bz(t) - cy(t). \end{aligned}$$

In matrix form, it is

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} y(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix}\begin{pmatrix} y(t) \\ z(t) \end{pmatrix}$$

Vector-valued ODEs and IVPs

Definition (Vector-valued first-order linear ODEs)

- Let $\boldsymbol{y}(t)$ and $\boldsymbol{b}(t)$ be vectors of length n whose elements are *functions*
- Let $\mathbf{A}(t)$ be an $n \times n$ matrix whose elements are also *functions*
- Then

$$\boldsymbol{y}'(t) = \mathbf{A}(t)\boldsymbol{y}(t) + \boldsymbol{b}(t)$$

is a vector-valued first-order linear ODE

Definition (IVP)

The corresponding IVP for a vector-valued first order ODE has initial condition

$$\boldsymbol{y}(t_0) = \boldsymbol{y}_0$$

where $\boldsymbol{y}_0$ is a vector of initial values.

Connection to linear algebra

- $\mathsf{T} = \dfrac{\mathrm{d}}{\mathrm{d}t} - \mathbf{A}\cdot$ is a *linear transformation* which operates on *vector-valued functions*.
- General solution is a complementary solution ($\boldsymbol{b} = \mathbf{0}$) plus a particular solution.

Theorem (Solution space of vector-valued first order linear ODEs)

- If $\mathbf{A}$ is continuous, the solution space has dimension n.
- A vector-valued first-order linear IVP has a unique vector-valued solution which is differentiable for all t on any interval in which $\mathbf{A}$ and $\boldsymbol{b}$ are continuous.

The Wronskian

Definition

- The Wronskian for $\boldsymbol{y}_1(t), \ldots, \boldsymbol{y}_n(t)$ is

$$\mathrm{wr}[\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n](t) = \det \left(\boldsymbol{y}_1 \middle| \ldots \middle| \boldsymbol{y}_n \right)$$

- If $\exists\, t$: $\mathrm{wr}[\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n](t) \neq 0$ then $\boldsymbol{y}_1(t), \ldots, \boldsymbol{y}_n(t)$ are linearly independent.
  - If any of $\boldsymbol{y}_1(t), \ldots, \boldsymbol{y}_n(t)$ are linearly dependent then $\mathrm{wr}[\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n](t) = 0$ for all t.
  - $\mathrm{wr}[\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n](t) = 0$ for some $t \not\Rightarrow \boldsymbol{y}_1(t), \ldots, \boldsymbol{y}_n(t)$ are linearly dependent.

Constant coefficients

Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$.

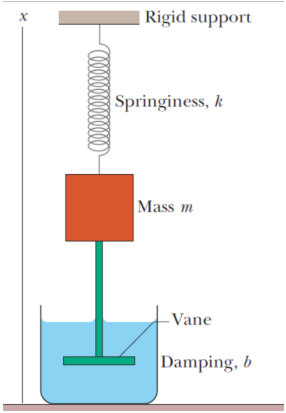
- Trial exponential solutions $\boldsymbol{y} = \boldsymbol{v}\,\mathrm{e}^{\lambda t}$ where $\lambda \in \mathbb{C}$ and $\boldsymbol{v} \in \mathbb{C}^n$. $\boldsymbol{y}' = \lambda\,\boldsymbol{v}\mathrm{e}^{\lambda t}(= \lambda\,\boldsymbol{y})$
- $\lambda \boldsymbol{v}\mathrm{e}^{\lambda t} = \mathbf{A}\boldsymbol{v}\mathrm{e}^{\lambda t}$ and so $\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}$

Recall linear algebra: *eigenvalues and eigenvectors*.

- Solve $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$ to find the eigenvalues, λ_i
- Find the corresponding eigenvectors, $\boldsymbol{v}_i$
- The eigenvectors form a basis for the solution space

Case study 1: Making vehicle platoons safer (Problem 1.2)

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 0, \quad x(0) = x_0, \quad x'(0) = v_0$$



Recall, the solution from revision tutorial

$$x(t) = (v_0 + 2x_0)\mathrm{e}^{-t} - (x_0 + v_0)\mathrm{e}^{-2t}$$

Sweeping through all possible initial conditions gives the solution space

‘Linear algebrise it’

Rewrite ODE as (Slide 7)

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} x \\ v \end{pmatrix} = \mathbf{A}\begin{pmatrix} x \\ v \end{pmatrix} \qquad \text{where} \qquad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} x(0) \\ v(0) \end{pmatrix} = \begin{pmatrix} x_0 \\ v_0 \end{pmatrix}$$

By trialling exponential solutions, solve the system of ODEs and establish linear independence of the solutions.

The eigenvalues of $\mathbf{A}$ are found from

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0 \quad \Rightarrow \quad (\lambda + 1)(\lambda + 2) = 0 \quad \Rightarrow \quad \lambda_1 = -1 \quad \text{and} \quad \lambda_2 = -2$$

The corresponding eigenvectors are

$$\boldsymbol{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \text{and} \quad \boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

So call

$$\boldsymbol{y}_1 = \mathrm{e}^{-t}\begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \text{and} \quad \boldsymbol{y}_2 = \mathrm{e}^{-2\,t}\begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Calculate the Wronskian:

$$\operatorname{wr}\left[\boldsymbol{y}_1, \boldsymbol{y}_2\right](t) = \det \begin{pmatrix} -\mathrm{e}^{-t} & -\mathrm{e}^{-2t} \\ \mathrm{e}^{-t} & 2\mathrm{e}^{-2t} \end{pmatrix} = -\mathrm{e}^{-3t}$$

Thus, the solutions are linearly independent and form a basis for the solution space.

Write a solution to the ODE as

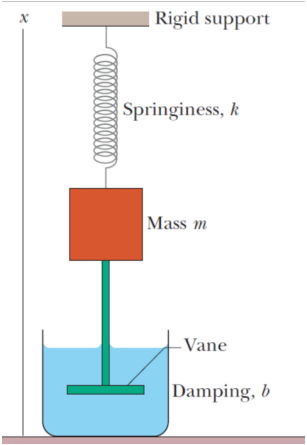
$$\boldsymbol{y} = \begin{pmatrix} x \\ v \end{pmatrix} = C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \mathrm{e}^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Applying the initial conditions gives $\begin{pmatrix} x_0 \\ v_0 \end{pmatrix} = C_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

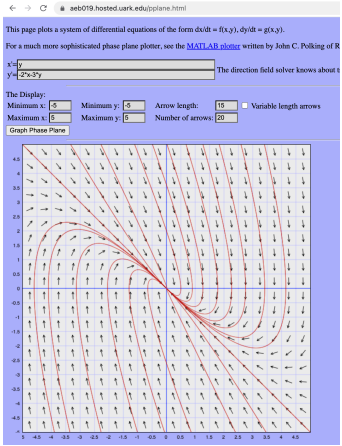
Solve the 2×2 system (exercise) to give $C_1 = -2x_0 - v_0$ and $C_2 = x_0 + v_0$

How does linear algebra dictate structure of solution space?

Physical space



Abstract phase space



Right-hand figure is from a webpage where you can plot phase portraits

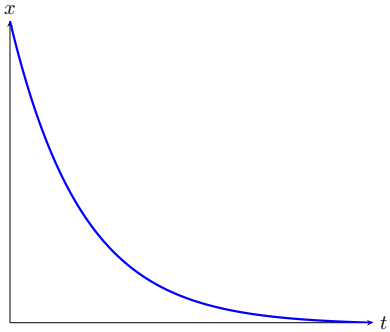
2nd order constant coefficient systems

The mass–spring system is a canonical example of a 2nd-order constant coefficient ODE

$$m\,x'' + b\,x' + k\,x = 0 \quad \text{where} \quad m, b, k > 0$$

Overdamped case

$$x(t) = C_1\,\mathrm{e}^{-A\,t} + C_2\,\mathrm{e}^{-B\,t}$$



We examine the overdamped case and leave the other cases for self-directed learning.

Exercise 1.12: Which case is the system from Problem 1.2?

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 2x = 0$$

$$\text{or} \qquad \boldsymbol{y}' = \mathbf{A}\boldsymbol{y} \quad \text{where} \quad \boldsymbol{y} = \begin{pmatrix} x \\ v \end{pmatrix} \quad \text{and} \quad \mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ v \end{pmatrix} = C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \mathrm{e}^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

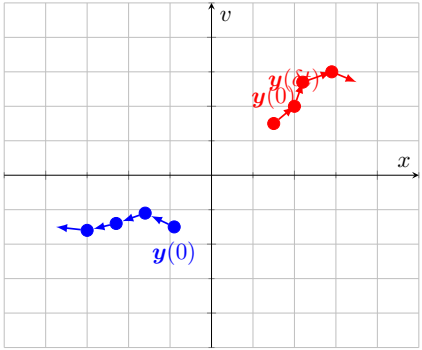
Real and distinct eigenvalues ($\lambda = -1$ and -2) = overdamped

- Fix C_1 and C_2 with ICs $x(0) = x_0$ and $x'(0) \equiv v(0) = v_0$ (or $\boldsymbol{y}(0) = \boldsymbol{y}_0$)
- Let's leave them unspecified for now.

We construct the phase portrait to establish the structure of solution space.

Phase space

- The xv-plane is called the **phase space**
- Seek solution as a planar parametric curve in t, called the **trajectory** or **orbit**
- Consider phase point $\boldsymbol{y}(0)$, representing the state of the system at time $t = 0$.
- $\boldsymbol{y}'(0) = \mathbf{A}\boldsymbol{y}(0)$ points along the trajectory
- Trace out trajectory by repeating this process
- Different $\boldsymbol{y}(0)$ leads to different trajectory
- Uniqueness of IVPs implies that trajectories cannot cross.

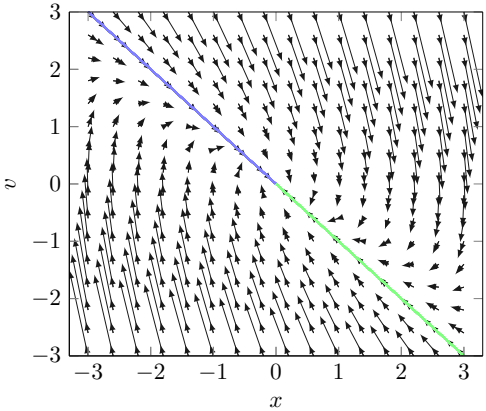


Vector field

- Plot vectors $\boldsymbol{y}'$ at mesh of points in phase space
- For Problem 1.2, plot

$$\begin{pmatrix} v \\ -2\,x - 3\,v \end{pmatrix}$$

  for different points (x, v) in phase space.
- Can help visualisation to normalise vector lengths, i.e., make vectors all same length



Critical points (or stationary points or equilibria)

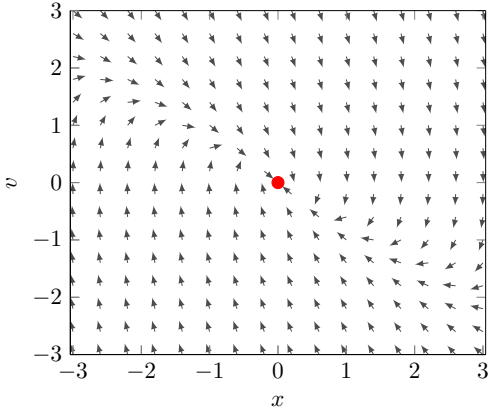
- Phase points $\boldsymbol{y}(t)$ such that $\mathbf{A}\boldsymbol{y} = \mathbf{0}$ are **critical points** of $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$.
- At critical points $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y} = \mathbf{0}$, and thus the solution does not change in time.
- In our example (Prb 1.2), we find $\boldsymbol{y}(t)$ such that

$$\underbrace{\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}}_{\mathbf{A}} \boldsymbol{y} = \mathbf{0} \quad \Rightarrow \quad \mathbf{y} = \mathbf{0}$$

- Only critical point of linear system $\boldsymbol{y}' = \mathbf{A}\boldsymbol{y}$ for $\mathbf{A}$ invertible is $\boldsymbol{y} = \mathbf{0}$

Phase portraits

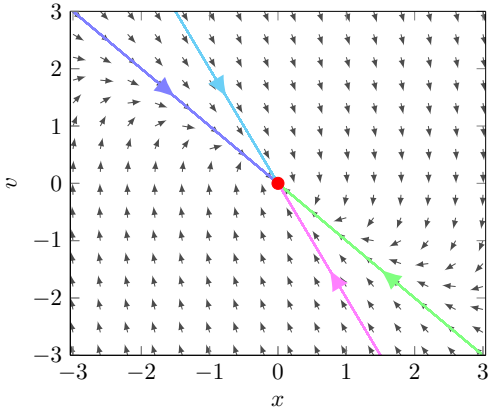
- *Phase portrait* is a graphical representation of the solution space
- Contains all critical points
- Contains representative orbits of system
- Matrix **A** alone shapes the structure of the solution space



Sketching phase portraits: Straight line orbits

What happens if IC is eigenvector of $\mathbf{A}$?

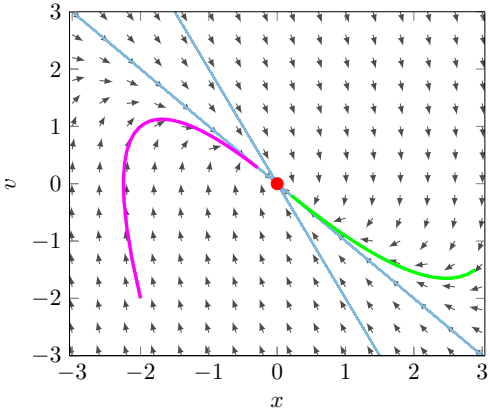
- Then $\mathbf{A}\boldsymbol{y}_0 = \lambda \boldsymbol{y}_0$
- So $\boldsymbol{y}'$ maintains its direction
  - Except possibly for a flip
- Orbit is straight line pointing in direction of increasing t
- For Prb 1.2
  - One eigenvector is $(-1, 1)^T$
  - Another is $(-1, 2)^T$



Sketching phase portraits - curved orbits

What happens if the initial condition is NOT along one of the eigenvectors?

- Linear transformation stretches/compresses/flips AND rotates initial condition.
- Solution is a parametric curve, with arrow pointing in direction of increasing t.
- Solution for large times 'grounded' by eigenvectors.



Sketching phase portraits - curved orbits

Back to Problem 1.2: Recall that

$$\boldsymbol{y}(t) = C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \mathrm{e}^{-2t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

- Take an IC

$$\Rightarrow \quad \boldsymbol{y}_0 = C_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad \text{for} \quad C_1, C_2 \neq 0$$

- For large t

$$\boldsymbol{y}(t) \approx C_1 \mathrm{e}^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

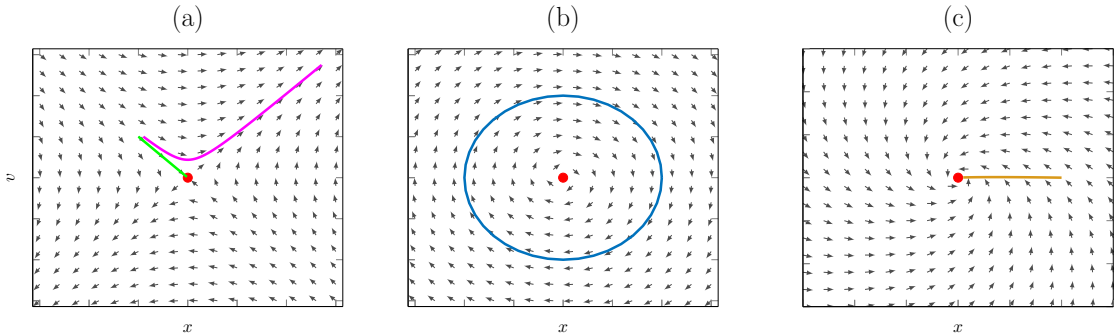
- This tell us that curved trajectories approach critical point along the first eigenvector as $t \to \infty$.

Sketching phase portraits: Stability of critical points

Study of large time behaviour is related to the **stability** of the critical points.

- A critical point is stable if, whenever a solution starts close enough to the critical point, it stays close forever for $t > 0$.
- A critical point is unstable if it is not stable.
  - This means that it is possible for a solution to start arbitrarily close to the critical point and eventually leave the neighbourhood of that point.
- The origin $\boldsymbol{y} = \mathbf{0}$ is stable if the real parts of the eigenvalues are negative or zero. In particular, it is asymptotically stable if the real parts of the eigenvalues are negative.

Exercise 1.13: Which of the following critical points is stable?



Diagonalisable systems

Revision

If $\mathbf{A}$ be a diagonalisable matrix, then $\mathbf{A} = \mathbf{S}\,\mathbf{D}\,\mathbf{S}^{-1}$ where

- $\mathbf{D}$ is a diagonal matrix whose entries are the eigenvalues
- and the columns of $\mathbf{S}$ contain the eigenvectors.

$$
\begin{aligned}
\boldsymbol{y}' &= \mathbf{A}\boldsymbol{y} + \boldsymbol{b} \\
\boldsymbol{y}' &= \mathbf{S}\mathbf{D}\mathbf{S}^{-1}\boldsymbol{y} + \boldsymbol{b} \\
\mathbf{S}^{-1}\boldsymbol{y}' &= \mathbf{D}\mathbf{S}^{-1}\boldsymbol{y} + \mathbf{S}^{-1}\boldsymbol{b} \\
\boldsymbol{z}' &= \mathbf{D}\boldsymbol{z} + \mathbf{S}^{-1}\boldsymbol{b} \quad \text{where} \quad \boldsymbol{z} = \mathbf{S}^{-1}\boldsymbol{y}.
\end{aligned}
$$

- We have decoupled the system.
- It can be solved as n independent first-order linear ODEs.

Exercise 1.14

Use the method of diagonalisation to solve the following system of ODEs:

$$x' - y = -\mathrm{e}^{-t}$$
$$y' + 2x + 3y = \mathrm{e}^{-t}$$

First we write as a vector equation

$$\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \boldsymbol{y} + \begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix}.$$

The eigenvalues and eigenvectors of the matrix are

$$\lambda_1 = -1 \quad \text{with} \quad \boldsymbol{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \qquad \text{and} \qquad \lambda_2 = -2 \quad \text{with} \quad \boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Thus

$$\boldsymbol{S} = \begin{pmatrix} -1 & -1 \\ 1 & 2 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{S}^{-1} = \begin{pmatrix} -2 & -1 \\ 1 & 1 \end{pmatrix}$$

The diagonalised system is

$$\boldsymbol{z}' = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \boldsymbol{z} + \boldsymbol{S}^{-1} \begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \boldsymbol{z} + \begin{pmatrix} \mathrm{e}^{-t} \\ 0 \end{pmatrix}$$

Solve this decoupled system using integrating factors (exercise) to get

$$\boldsymbol{z} = \begin{pmatrix} (C_1 + t)\,\mathrm{e}^{-t} \\ C_2 \mathrm{e}^{-2t} \end{pmatrix}$$

$$\Rightarrow \quad \boldsymbol{y} = \boldsymbol{S}\boldsymbol{z} = \begin{pmatrix} -(C_1 + t)\,\mathrm{e}^{-t} - C_2 \mathrm{e}^{-2t} \\ (C_1 + t)\,\mathrm{e}^{-t} + 2C_2 \mathrm{e}^{-2t} \end{pmatrix}$$

Summary of systems ODEs

1. Higher order ODEs can be written as a single *vector-valued* ODE.
2. Constant coefficient homogeneous: use exponential trial functions.
3. Constant coefficient inhomogeneous: use matrix diagonalisation to decouple the system then use an integrating factor on individual ODEs.

When does it fall apart?

- Matrix is not diagonalisable or coefficients are functions of t: learn matrix exponentials!